Chapter

User Defined Methods

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1. Assertion (A) : Non-member methods are tied to specific objects in Java.
Reason (R) : Non-member methods are declared using the static keyword and can be accessed without creating an object. **SQP 2024**

(a) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.

2. What is an advantage of using functions in programming? MAIN 2024

(a) Code repetition
(b) Abstraction
(c) Increased complexity
(d) Slower debugging

3. What does the term "function signature" refer to? SQP 2023

(a) The body of the function
(b) The return type of the function
(c) The combination of the functions name and parameter list
(d) The access specifier of the function

4. What is the correct syntax for creating an object in Java? COMP 2022

(a) ClassName objectName = ClassName();
(b) ObjectName = new ClassName();
(c) ClassName objectName = new ClassName();
(d) New ClassName objectName;

5. Which of the following is an example of a nonmember method? MAIN 2021

(a) Void display()
(b) Static void checkSum()
(c) Void sum()
(d) Public void calculate()

6. A student writes the following code to double the value of a variable q. He has written the following statement, which is incorrect: COMP 2023
q + q;
What will be the correct statement?
A. q *= 2;
B. q = q * 2;
C. q = q + q;

(a) Only A
(b) Only B
(c) Both A and B
(d) All A, B, and C

7. (i) Write two advantages of using functions in a program
(ii) Differentiate between static and non-static data members **MAIN 2024**

8. Write a function prototype of the following . COMP 2021
(i) A function PosChar which takes a string argument and a character argument and returns an integer value
(ii) Function that returns a double value and takes three double parameters x, y, and z as its parameters.

9. Write a function prototype of the following. MAIN 2022
(i) Function that returns a boolean value and takes an integer variable as its argument
(ii) A static method named calculator that accepts two short type variables as its arguments and returns an integer value.

10. Write a function prototype of the following COMP 2024

(i) A function divide that takes two integer values and returns the quotient of double type.
(ii) An overloaded function that takes two double values as formal parameters and returns the quotient of double type

11. Write the general format of defining a function in a java class SQP 2023

12. Define a Java method temp-convert to convert Fahrenheit temperature to Celsius. The formula for the conversion is given as SQP 2024

$C = \frac{5}{9}(F - 32), for\ F \text{ taken as } 45^\circ F.$

13. Write a complete Java program that contains the following methods assign() : Accepts values of the digit at units place, the digit at tens place, and the digit at hundreds place
findNumber() : Finds the number corresponding to the accepted digits main() : Calls the above methods and displays the computed number COMP 2024

Solution

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1. (d) Assertion (A) is false, but Reason (R) is true.

Non-member methods in Java are declared as **static**, so they belong to the class and **can be accessed without creating an object**. Therefore, the reason is true. The assertion that non-member methods are tied to specific objects is false.

2. (b) Abstraction

Functions **abstract the complexity of a program** by grouping a set of instructions into a single, reusable block. This reduces redundancy, improves code clarity, and makes the program easier to maintain and debug.

3. (c) The combination of the function's name and parameter list

A **function signature** uniquely identifies a function in a program. It includes the **function's name** and the **types and number of its parameters**, which helps in distinguishing between overloaded functions.

4. (c) ClassName objectName = new ClassName();

In Java, an object is created by using the new keyword followed by the class constructor. Here, ClassName is the class, and objectName is the reference variable that points to the newly created object.

5. (b) Static void checkSum()

A non-member method in Java is a **static method**. It belongs to the class rather than any specific object and can be called without creating an instance of the class. Here, checkSum() is declared as static, making it a non-member method.

6. (d) All A, B, and C

The student's statement q + q; is incorrect because it only calculates the sum but does not store it. The correct ways to double q are:

1. q *= 2;

2. q = q * 2;

3. q = q + q;

All three properly assign the doubled value back to q.

7. (i) Advantages of using functions:

- **Reusability of code:** Functions allow a block of code to be written once and used multiple times.
- **Easy debugging and testing:** Functions break the program into smaller modules, making it easier to test and debug.

(ii) Difference between static and non-static data members:

- **Static members:** Belong to the class itself, shared by all objects, and can be accessed without creating an object.
- **Non-static members:** Belong to individual objects, and each object has its own copy of these members.

8. (i) int PosChar(String st, char a);
(ii) double funcQ7(double x, double y, double z);

9. (i) boolean funcQ8(int x);
(ii) static int calculator(short x, short y);

10. (i) double divide(int x, int y);
(ii) double divide(double m, double n);

11. [access specifier] [static] returntype
functionname (formal parameters)
{
statements; // Function body
}

12.

```
public class Temp {
              public    static    double
tempConvert(float Fahren) {
            double cel = 5.0 / 9.0 *
(Fahren - 32);
         return cel;
     }

     public static void main() {
                              System.out.
println(tempConvert(45));
     }
}
```

13.

```
import java.util.*;
public class Q15 {
      private static int unit, tens,
hun;

     static void assign(int u, int t,
int h) {
          unit = u;
          tens = t;
          hun = h;
     }

     static long findNumber() {
           return (hun * 100 + tens *
10 + unit);
     }

      public static void main(String
args[]) {
      Scanner buf = new Scanner(System.
in);
            System.out.println("Please
enter a units place, the tens place
and hundreds place digits : ");
          int n1 = buf.nextInt();
          int n2 = buf.nextInt();
          int n3 = buf.nextInt();
          assign(n1, n2, n3);
        System.out.println("The number
is : " + findNumber());
     }
}
```